Trading Results Audit: Strategies, Pairs, and Paper Coverage

DeFi Long–Short / Carry Framework (`Defi_interestRate`)

Prepared as a coverage and sanity check of the empirical trading work

2026-09-13

Abstract

This note answers three questions about the trading work in the repository: (i) *are the trading results OK?*, (ii) *which strategies were tried, on which pairs?*, and (iii) *does the final paper* (`latex/optimal_long_short.tex`) *include the most relevant use case(s)?* The short answers: the results are internally coherent and deliver a clean, honest *negative* result (the naive leveraged carry book is a bad trade). The Sharpe convention has now been fixed (Section 1): the benchmark previously scored every strategy against a hardcoded flat 5% and mixed an arithmetic Sharpe numerator with a geometric Sortino numerator. After correcting both — risk-free set to the realised on-chain USDC supply rate, and Sharpe/Sortino/Omega placed on one consistent excess series — the Sharpe ratios stay negative, but now for a *genuine economic reason*: the mean USDC supply rate over the window was 4.80%, and *no* strategy beat simply supplying USDC. Eight distinct strategies were run across a 37-pair cross-section, with almost all backtesting on `USDC`(supply)/`WETH`(borrow) and the analytics on `WBTC`/`WETH`. The final paper contains the diagnostic use cases (sizing, carry-inert cross-section, regime switching, depeg, recovery) but **omits the single most actionable use case — the Taleb barbell** — which the historical benchmark identifies as the best strategy on a tail-risk basis. The paper's negative result is present; its constructive answer is not.

1 Are the trading results OK?

Verdict. *Directionally yes, with one methodological caveat that must be resolved before publication.* The results are consistent with the paper's thesis and with the pre-registered cross-section: the leveraged long–short / carry book on Aave is structurally short two convexities (the liquidation barrier and the utilisation kink) and collects negative net carry on every ETH/BTC-collateral pair, so it should — and does — lose money once liquidations are counted. That is a genuine, valuable finding, not a bug.

What is solid.

- **The negative result reproduces out of sample.** In the walk-forward file (`optimal_long_short/walk_forwar` 19 non-overlapping 45-day entries, 2023-11 → 2026-02) the naive carry leg is liquidated in 5/19 windows and posts large negative PnL in several more; the barbell and pure-supply legs stay small and bounded.
- **The cross-section is unambiguous.** In `results/pair_cross_section.csv`, *every* ETH- and BTC-collateral pair has negative net carry (roughly −1.8% to −4.5%/yr); only stable/stable

legs (e.g. `USDC/USDe`) show marginally positive carry. This is the structural premise of the whole framework and it holds across all 37 pairs.

- **The Monte-Carlo carry study is coherent.** `results/aave_carry_trade_metrics.csv` (CIR vs. CIR+Jump, 10 000 paths) gives ∼6.7–6.9% liquidation probability with ∼91–96% of surviving paths profitable — a sensible "small edge, fat left tail" shape.

The Sharpe convention (now fixed — see Section 1).

- **Every strategy has a negative Sharpe ratio** in the 5-strategy historical benchmark (Table 1), *including* strategies with positive cumulative return (the barbell: +15.1% cumulative, +4.3% annualised). The original code had two defects: it benchmarked against a flat hard-coded 5% risk-free, and it computed the Sharpe numerator arithmetically while computing the Sortino numerator geometrically. Both are fixed. The surprise: the correction *does not* flip the signs — the realised mean USDC supply rate was 4.80%, so the flat 5% was nearly right, and the negative Sharpes are economically real: **no leveraged strategy beat the ∼4.8% risk-free of simply supplying USDC**. This is a sharper finding than the original, not a bug artifact. (Sortino moved materially, e.g. barbell $-0.65 \to -0.08$, because the inconsistent numerator is gone.)
- **The carry blow-up is a total wipeout** (−100% cumulative, Sharpe −3.31, max drawdown −100%). That is a knockout artifact (equity hits zero once, path ends), which is *correct* as a statement that the trade can be ruined, but it should be reported as "ruin probability / time to ruin," not as a return series, since a −100% return dominates every summary statistic it enters.
- **Sign conventions on the risk columns are inconsistent** (e.g. `VaR 95%` is reported as −5.44 for the barbell, i.e. a gain, but as +5886 for carry). Standardise before publishing.

Table 1: Historical benchmark, Aave V3 Ethereum, Jan 2023 – Jun 2026 (1,222 daily obs, $100k). Source: `strategy_benchmark_executed.ipynb`, **Sharpe/Sortino recomputed with the corrected convention** (rf = on-chain USDC supply, 4.80% mean). The negative Sharpes are real: no strategy beats the risk-free supply rate; the barbell/momentum win only on tail-risk (Calmar).

Strategy	Cum. Ret.	Ann. Ret.	Ann. Vol.	Sharpe	Sortino	Max DD	Calmar
Current Carry Trade	−100.0%	−100.0%	224.8%	−3.31	−1.20	−100.0%	−1.00
Taleb Barbell	+15.1%	+4.3%	1.1%	−0.40	−0.08	−2.6%	+1.66
Pure Supply & Demand	−73.8%	−33.0%	44.9%	−0.76	−0.44	−79.6%	−0.41
Black–Litterman	−4.1%	−1.2%	25.5%	−0.10	−0.09	−27.0%	−0.05
Momentum & Factor Rotation	+13.8%	+3.9%	2.9%	−0.25	−0.33	−3.1%	+1.28

The Sharpe convention fix

The metric helper (`compute_metrics` in `strategy_benchmark.ipynb`) was corrected in two ways, and the notebook re-executed:

1. **Risk-free = on-chain USDC supply rate.** The book is USDC-denominated, so the opportunity cost is what idle capital earns supplying USDC on Aave, not a flat exogenous 5%. A per-day `RF_DAILY` series is built from the realised supply APR (geometric annual→daily) and used as the excess-return base, aligned to each strategy's dates.

2. **Consistent numerators.** Sharpe, Sortino and Omega now use the *same* per-period excess series $r - r_f$ and the same arithmetic $\sqrt{365}$ annualisation. Previously Sharpe used an arithmetic excess mean while Sortino used a geometric (CAGR $- r_f$) numerator — an internal inconsistency that made the two ratios non-comparable.

Net effect: Sharpe barely moves (the flat 5% was coincidentally close to the 4.80% realised supply rate), Sortino moves materially, and the qualitative conclusion is unchanged and now defensible.
Bottom line. The economics are right and the negative result is real and well-supported. With the Sharpe convention now fixed, the honest reading is: over Jan 2023–Jun 2026 *no* leveraged strategy out-earned the ∼4.8% risk-free of just supplying USDC on a risk-adjusted basis; the barbell and momentum rotation are the only two that preserve capital (positive Calmar, bounded drawdown) and are the sensible choices for a tail-averse allocator. The one remaining cosmetic item is the VaR sign convention (negative = gain for the near-riskless strategies), which is confusing but not a bug.

2 Strategies tried and on which pairs

Analytic / model use cases (in the framework)

Use case	Where	Pair(s)
Unit-exposure long–short sizing (survival prob. + killed moments)	`strategy.py`, `risk_report.py`	`WBTC` coll. / `WETH` debt
Carry-adjusted sizing ($\mu_i \to \mu_i + r$)	`carry.py`	Any Aave pair; headline `WBTC/WETH`
Carry cross-section (deterministic carry inert)	`screen.py`, `results/pair_cross_section.csv`	**37 ordered pairs** over {USDC, USDT, USDe, WBTC, WETH, cbBTC, weETH, wstETH}
Regime-switching carry (6×6 resolvent)	`regime_switching.py`	`WBTC/WETH`
Collateral-depeg sensitivity	`jobs/*`, `usdc_depeg_sensitivity.png`	Stable-collateral legs
Liquidation recovery / bad-debt	`recovery.py`	`WBTC/WETH`

Backtested / simulated trading strategies

Strategy	Mechanism	Pair(s) / universe
1. Current carry trade	`USDC` supply + `WETH` borrow, LTV 70%	`USDC/WETH`
2. Taleb barbell	90% safe supply + 10% carry; supplier + keeper convexity legs	`USDC/WETH` (`barbell.py`)
3. Pure supply & demand	On-chain utilisation signal, LTV 50%	`USDC` (supply)

Strategy	Mechanism	Pair(s) / universe
4. Black–Litterman	TVL-weighted Bayesian carry, weekly rebalance	USDC, DAI, USDT, WETH
5. Momentum & factor rotation	4 assets × 5 factors, weekly top-2	USDC, DAI, USDT, WETH
6. CIR / CIR+Jump carry MC	10k-path rate simulation, liq. stats	USDC/WETH
7. Walk-forward OOS	carry vs. supply vs. barbell, 45-day rolls	USDC/WETH

Pair coverage summary. Model analytics: headline WBTC/WETH plus a full 37-pair Aave cross-section. Trading backtests: almost entirely USDC(supply)/WETH(borrow), with DAI/USDT added only as diversifiers in strategies 4–5. *No backtest is run on the BTC-collateral pairs that the paper analyses*, so the analytic headline pair and the traded pair differ — worth reconciling.

3 Does the final paper include the most relevant use case(s)?

The final paper is `latex/optimal_long_short.tex`. Its sections cover: Introduction; Problem Setup; Asset Dynamics (Kou); Moments via Measure Change; Resolvent Equations; Numerical Results + sensitivity; Liquidation Recovery; Carry-Adjusted Sizing (incl. the inert cross-section, the regime-switching 6×6 extension, and collateral-depeg); Conclusion (the negative, "no interior optimum" result); and the ECF calibration appendix.

Table 4: Use-case coverage in the final paper.

Use case	In optimal_long_short.tex?	Note
Long–short sizing / survival / moments	yes	Core of the paper
Carry as effective drift	yes	§ "Carry-Adjusted Sizing"
Carry cross-section (carry inert)	yes	37 pairs
Regime-switching carry (6×6)	yes	
Collateral-depeg sensitivity	yes	
Liquidation recovery / bad debt	yes	Optimistic bound (see critique)
The negative result (no interior optimum)	yes	The paper's best feature
Taleb barbell (constructive answer)	no	**Only in code/notebook**
5-strategy historical benchmark	no	Only in notebook
Walk-forward OOS validation	no	Only in CSV
CIR/JCIR carry Monte Carlo	no	Only in CSV/plots

The most relevant use case is missing. The paper's central thesis is a *negative* one: the leveraged long–short book is structurally a bad trade. The **barbell** (`barbell.py`) is the direct constructive response — it longs exactly the two convexities the levered book is short (the barrier via a keeper leg, the utilisation kink via a supplier leg) — and the historical benchmark ranks it *first* on Calmar (1.66) and tail metrics, with a bounded -2.6% max drawdown versus the carry trade's -100%. It is the single most actionable output of the whole project, and it appears *only* in code and an executed notebook, never in the paper. This mirrors the standing critique in `CRITICAL_REPORT.md`

§3.4: the paper diagnoses that mean–variance gives no interior optimum but never evaluates the growth-optimal / convexity-harvesting alternative that would.

Recommendation. Add one section to `optimal_long_short.tex` — "From the negative result to a tradable barbell" — that (i) states the two short-convexity exposures formally, (ii) prices the supplier and keeper legs with the existing first-passage transform, and (iii) reports the tail-risk-dominant backtest, *after* fixing the Sharpe/VaR convention bug flagged in Section 1. Without it, the paper delivers the diagnosis but withholds the cure that the codebase already computes.